MATHS 351

Assignment 1

Question 1

a. $\begin{pmatrix} 6 & -2 & 2 & 4 \\ 12 & -8 & -6 & 10 \\ 3 & -13 & 9 & 3 \\ -6 & 4 & 1 & -18 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 16 \\ 26 \\ -19 \\ -34 \end{pmatrix}$

Comparing to $Ax = b$; $A = \begin{pmatrix} 6 & -2 & 2 & 4 \\ 12 & -8 & -6 & 10 \\ 3 & -13 & 9 & 3 \\ -6 & 4 & 1 & -18 \end{pmatrix}$; $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$; $b = \begin{pmatrix} 16 \\ 26 \\ -19 \\ -34 \end{pmatrix}$

b. Using Naive Gaussian Elimination method

$M_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ -1/2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$; using the matrix approach, the Augmented matrix becomes

$A_1 = \left(\begin{array}{cccc|c} 6 & -2 & 2 & 4 & 16 \\ 12 & -8 & -6 & 10 & 26 \\ 3 & -13 & 9 & 3 & -19 \\ -6 & 4 & 1 & -18 & -34 \end{array}\right)$; $M_1 A_1 = A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ -1/2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \left(\begin{array}{cccc|c} 6 & -2 & 2 & 4 & 16 \\ 12 & -8 & -6 & 10 & 26 \\ 3 & -13 & 9 & 3 & -19 \\ -6 & 4 & 1 & -18 & -34 \end{array}\right)$

$A_2 = \left(\begin{array}{cccc|c} 6 & -2 & 2 & 4 & 16 \\ 0 & -4 & -10 & 2 & -6 \\ 0 & -12 & 8 & 1 & -27 \\ 0 & 2 & 3 & -14 & -18 \end{array}\right)$; $M_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -3 & 1 & 0 \\ 1 & 1/2 & 0 & 1 \end{pmatrix}$

$M_2 A_2 = A_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -3 & 1 & 0 \\ 1 & 1/2 & 0 & 1 \end{pmatrix} \left(\begin{array}{cccc|c} 6 & -2 & 2 & 4 & 16 \\ 0 & -4 & -10 & 2 & -6 \\ 0 & -12 & 8 & 1 & -27 \\ 0 & 2 & 3 & -14 & -18 \end{array}\right) = \left(\begin{array}{cccc|c} 6 & -2 & 2 & 4 & 16 \\ 0 & -4 & -10 & 2 & -6 \\ 0 & 0 & 38 & -5 & -9 \\ 0 & 2 & -2 & -13 & -21 \end{array}\right)$

$M_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 1/19 & 1 \end{pmatrix}$; $M_3 A_3 = A_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -3 & 1 & 0 \\ 1 & 1/2 & 0 & 1 \end{pmatrix} \left(\begin{array}{cccc|c} 6 & -2 & 2 & 4 & 16 \\ 0 & -4 & -10 & 2 & -6 \\ 0 & -12 & 8 & 1 & -27 \\ 0 & 2 & 3 & -14 & -18 \end{array}\right)$

$A_4 = \left(\begin{array}{cccc|c} 6 & -2 & 2 & 4 & 16 \\ 0 & -4 & -10 & 2 & -6 \\ 0 & 0 & 38 & -5 & -9 \\ 0 & 0 & 0 & -252/19 & -408/19 \end{array}\right)$

$$\therefore \begin{pmatrix} 6 & -2 & 2 & 4 \\ 0 & -4 & -10 & 2 \\ 0 & 0 & 38 & -5 \\ 0 & 0 & 0 & -252/19 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 16 \\ -6 \\ -9 \\ -408/19 \end{pmatrix}$$; employing back substitution,

$$\frac{-252x_4}{19} = \frac{-408}{19} \Rightarrow x_4 = \frac{34}{21}$$

$$38x_3 - 5x_4 = -9 \Rightarrow x_3 = \frac{-9 + 5\left({}^{34}/_{21}\right)}{38} \Rightarrow x_3 = \frac{-1}{42}$$

$$-4x_2 - 10x_3 + 2x_4 = -6 \Rightarrow x_2 = \frac{-6 + 10\left({}^{-1}/_{42}\right) - 2\left({}^{34}/_{21}\right)}{-4} \Rightarrow x_2 = \frac{199}{84}$$

$$6x_1 - 2x_2 + 2x_3 + 4x_4 = 16 \Rightarrow x_1 = \frac{16 + 2\left({}^{199}/_{84}\right) - 2\left({}^{-1}/_{42}\right) - 4\left({}^{34}/_{21}\right)}{6} \Rightarrow x_1 = \frac{601}{252}$$

$$\therefore \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 601/252 \\ 199/84 \\ -1/42 \\ 34/21 \end{pmatrix}$$

Using Gaussian Elimination with partial pivoting

$A_1 = \left(\begin{array}{cccc|c} 6 & -2 & 2 & 4 & 16 \\ 12 & -8 & -6 & 10 & 26 \\ 3 & -13 & 9 & 3 & -19 \\ -6 & 4 & 1 & -18 & -34 \end{array}\right)$; for the 1st forward elimination, $|6|, |12|, |3|\ and\ |-6|$, the largest value is in R1. Hence R1 is switched with R2 to give;

$$A_1 = \left(\begin{array}{cccc|c} 12 & -8 & -6 & 10 & 26 \\ 6 & -2 & 2 & 4 & 16 \\ 3 & -13 & 9 & 3 & -19 \\ -6 & 4 & 1 & -18 & -34 \end{array}\right);\ M_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1/2 & 1 & 0 & 0 \\ -1/4 & 0 & 1 & 0 \\ 1/2 & 0 & 0 & 1 \end{pmatrix}$$

$$M_1 A_1 = A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1/2 & 1 & 0 & 0 \\ -1/4 & 0 & 1 & 0 \\ 1/2 & 0 & 0 & 1 \end{pmatrix} \left(\begin{array}{cccc|c} 12 & -8 & -6 & 10 & 26 \\ 6 & -2 & 2 & 4 & 16 \\ 3 & -13 & 9 & 3 & -19 \\ -6 & 4 & 1 & -18 & -34 \end{array}\right)$$

$A_2 = \left(\begin{array}{cccc|c} 12 & -8 & -6 & 10 & 26 \\ 0 & -2 & 5 & -1 & 3 \\ 0 & -11 & 21/2 & 1/2 & -51/2 \\ 0 & 0 & -2 & -13 & -21 \end{array}\right)$; for the 2nd forward elimination, $|2|, |-11|\ and\ |0|$, the largest value is in R3, therefore R3 is swapped with R2 to give;

$$A_2 = \left(\begin{array}{cccc|c} 12 & -8 & -6 & 10 & 26 \\ 0 & -11 & 21/2 & 1/2 & -51/2 \\ 0 & 2 & 5 & -1 & 3 \\ 0 & 0 & -2 & -13 & -21 \end{array}\right);\ M_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 2/11 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$M_2A_2 = A_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 2/11 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \left(\begin{array}{cccc|c} 12 & -8 & -6 & 10 & 26 \\ 0 & -11 & 21/2 & 1/2 & -51/2 \\ 0 & 2 & 5 & -1 & 3 \\ 0 & 0 & -2 & -13 & -21 \end{array} \right)$$

$$A_3 = \left(\begin{array}{cccc|c} 12 & -8 & -6 & 10 & 26 \\ 0 & -11 & 21/2 & 1/2 & -51/2 \\ 0 & 0 & 76/11 & -10/11 & -18/11 \\ 0 & 0 & -2 & -13 & -21 \end{array} \right)$$; for the 3rd forward elimination, $\left|{}^{76}/_{11}\right| \; and \; |-2|$, the largest value is in R3, hence no swapping of Rows between R3 and R4.

$$M_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 11/38 & 1 \end{pmatrix};$$

$$M_3A_3 = A_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 11/38 & 1 \end{pmatrix} \left(\begin{array}{cccc|c} 12 & -8 & -6 & 10 & 26 \\ 0 & -11 & 21/2 & 1/2 & -51/2 \\ 0 & 0 & 76/11 & -10/11 & -18/11 \\ 0 & 0 & -2 & -13 & -21 \end{array} \right)$$

$$A_4 = \left(\begin{array}{cccc|c} 12 & -8 & -6 & 10 & 26 \\ 0 & -11 & 21/2 & 1/2 & -51/2 \\ 0 & 0 & 76/11 & -10/11 & -18/11 \\ 0 & 0 & 0 & -252/19 & -408/19 \end{array} \right)$$; employing back substitution,

$$\therefore \frac{-252x_4}{19} = \frac{-408}{19} \Rightarrow x_4 = \frac{34}{21}$$

$$\frac{76x_3}{11} - \frac{10x_4}{11} = \frac{-18}{11} \Rightarrow x_3 = \frac{-1}{42}$$

$$-11x_2 + \frac{21x_4}{2} + \frac{x_4}{2} = \frac{-54}{2} \Rightarrow x_2 = \frac{199}{84}$$

$$12x_1 - 8x_2 - 6x_3 + 10x_4 = 26 \Rightarrow x_1 = \frac{601}{252}$$

$$\therefore \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 601/252 \\ 199/84 \\ -1/42 \\ 34/21 \end{pmatrix}$$

c. The advantages of Gaussian Elimination with partial pivoting over Naïve Gaussian Elimination method are as follows;
   I. It helps correct or reduce the errors made in round-offs.
   II. It also helps get-rid of any possibility of encountering division by zero.

QUESTION 2

$\begin{pmatrix} 2.11 & -4.21 & 0.9 \\ 4.01 & 10.2 & -1.12 \\ 1.09 & 0.987 & 0.832 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2.01 \\ -3.09 \\ 4.21 \end{pmatrix}$; Augmented form of the matrix becomes

$A_1 = \left(\begin{array}{ccc|c} 2.11 & -4.21 & 0.9 & 2.01 \\ 4.01 & 10.2 & -1.12 & -3.09 \\ 1.09 & 0.987 & 0.832 & 4.21 \end{array}\right)$; Complete pivoting aims at making the largest element in the matrix your pivot element by the swapping both rows and columns unlike in partial pivoting where we only change rows.

Hence swapping C1 and C2, $A_1 = \left(\begin{array}{ccc|c} -4.21 & 2.11 & 0.9 & 2.01 \\ 10.2 & 4.01 & -1.12 & -3.09 \\ 0.987 & 1.09 & 0.832 & 4.21 \end{array}\right)$, In the swapping of columns, the coefficients of $x_1\ and\ x_2$ also interchanges.

$R1 \Leftrightarrow R2$, $A_1 = \left(\begin{array}{ccc|c} 10.2 & 4.01 & -1.12 & -3.09 \\ -4.21 & 2.11 & 0.9 & 2.01 \\ 0.987 & 1.09 & 0.832 & 4.21 \end{array}\right)$; $M_1 = \begin{pmatrix} 1 & 0 & 0 \\ 421/1020 & 1 & 0 \\ -329/3400 & 0 & 1 \end{pmatrix}$

$M_1A_1 = A_2 = \begin{pmatrix} 1 & 0 & 0 \\ 421/1020 & 1 & 0 \\ -329/3400 & 0 & 1 \end{pmatrix}\left(\begin{array}{ccc|c} 10.2 & 4.01 & -1.12 & -3.09 \\ -4.21 & 2.11 & 0.9 & 2.01 \\ 0.987 & 1.09 & 0.832 & 4.21 \end{array}\right)$

$A_2 = \left(\begin{array}{ccc|c} 10.2 & 4.01 & -1.12 & -3.09 \\ 0 & 3.765108 & 0.437725 & 0.734618 \\ 0 & 0.701975 & 0.940376 & 4.509003 \end{array}\right)$, for the next pivoting element, there wouldn't be any swapping since the largest element stands in the position of the 2nd principal minor.

$M_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -0.186442 & 1 \end{pmatrix}$

$M_2A_2 = A_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -0.186442 & 1 \end{pmatrix}\left(\begin{array}{ccc|c} 10.2 & 4.01 & -1.12 & -3.09 \\ 0 & 3.765108 & 0.437725 & 0.734618 \\ 0 & 0.701975 & 0.940376 & 4.509003 \end{array}\right)$

$A_3 = \left(\begin{array}{ccc|c} 10.2 & 4.01 & -1.12 & -3.09 \\ 0 & 3.765108 & 0.437725 & 0.734618 \\ 0 & 0 & 0.858766 & 4.372039 \end{array}\right)$

$\begin{pmatrix} 10.2 & 4.01 & -1.12 \\ 0 & 3.765108 & 0.437725 \\ 0 & 0 & 0.858766 \end{pmatrix}\begin{pmatrix} x_2 \\ x_1 \\ x_3 \end{pmatrix} = \begin{pmatrix} -3.09 \\ 0.734618 \\ 4.372039 \end{pmatrix}$, employing substitution,

$0.858766x_3 = 4.372039 \Rightarrow x_3 = 5.091071$

$3.765108x_1 + 0.437725x_3 = 0.734618 \Rightarrow x_1 = \dfrac{0.734618 - 0.437725x_3}{3.765108} \Rightarrow x_1 = -0.396767$

$10.2x_1 + 4.01x_1 - 1.12x_3 = -3.09 \Rightarrow x_2 = \dfrac{-3.09 - 4.01x_1 + 1.12x_3}{10.2} \Rightarrow x_2 = 0.412062$

$$\therefore \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -0.396767 \\ 0.412062 \\ 5.091071 \end{pmatrix}.$$